Review notes · for you

First Python: unit check

Programming · Level 2 · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Hello, Python
- Variables and input
- Decisions
- Loops

Activity: Quiz show

1. Warm-up (3 min): Write one thing you learned in each of the four lessons, in your own words.
2. Make (7 min): In pairs, write a quiz program on paper: it asks three questions with input(), uses if to check each answer, and keeps a score.
3. Swap (7 min): Swap with another pair. Predict what their program prints for some answers, and look for bugs.
4. Run (3 min): Type one quiz into the phone and play it. Did it behave as predicted?

Discussion

- Which lesson was the hardest, and why?
- Which error message do you understand best now?
- What program will you write next?

Check yourself

- One program I could write now that I couldn't before

Try it this week
Write one program for your own life that uses input, an if and a loop, such as a quiz with three questions that counts your score.

Quiz sheet

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Name _____ Class _____ Date _____

1 **What does this line show?**

```
print(10 - 2 * 3)
```

◯ 4 ◯ 24 ◯ 10 - 2 * 3

2 **input() always gives back text, even when you type a number.**

◯ True ◯ False

3 **Fill the gaps: the program says Adult for 18 or older.**

```
age = _____(input("Age? "))
if age _____ 18:
    print("Adult")
```

Word bank: `>=` `=` `int` `str`

4 **Python stops this program with an IndentationError. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  for i in range(3):
2  print("Hi")
```

---

5 **Match each line to what it prints.**

print("3" + "4")	• • • •	ababab
print(3 + 4)	• • • •	16
print(3 * "ab")	• • • •	34
print(2 ** 4)	• • • •	7

6 **Put the lines in order: the program counts down 3, 2, 1, then prints Go!. One line doesn't belong.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

- [ ] `print(n)`
- [ ] `while n < 0:`
- [ ] `n = 3`
- [ ] `print("Go!")`
- [ ] `n = n - 1`
- [ ] `while n > 0:`

7 **Free under 5, half price under 12, else full price. A boy is exactly 5. What does he pay?**

◯ Full price ◯ Free ◯ Half price

8 **Which loops print exactly 5 lines?**

☐ for i in range(5): ☐ for i in range(0, 5): ☐ for i in range(1, 6): ☐ for i in range(1, 5):

9 **Finish the program: it asks for two marks and prints the higher one. If they are equal, it prints that mark.**

```
a = int(input("First mark? "))
b = int(input("Second mark? "))
# print the higher mark
```

Write your program:

10 **if score = 10: is a correct line of Python.**

◯ True ◯ False

11 **Put the steps for fixing an error in order.**

- [ ] Go to the line it names
- [ ] Run it again
- [ ] Run the program and read the error message
- [ ] Fix one thing

Check your answers

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1 **4**

Python multiplies first: 2 * 3 = 6, then 10 - 6 = 4.

2 **True**

Typing 13 gives the text "13". Use int() to turn it into the number 13.

3

```
age = int(input("Age? "))
if age >= 18:
    print("Adult")
```

int() makes the answer a number so it can be compared, and >= includes 18 itself.

4 **Line 2: `print("Hi")`**

The lines inside a loop must be indented by 4 spaces, so Python knows they belong to it.

5

print("3" + "4") — 34
print(3 + 4) — 7
print(3 * "ab") — abababab
print(2 ** 4) — 16

Text + text joins; number + number adds; a number times text repeats it; ** is a power.

6 **Top to bottom: 3, X, 1, 5, 4, 2**

```
n = 3
while n > 0:
    print(n)
    n = n - 1
print("Go!")
```

The loop runs while n is still above 0, printing then going down by 1.

7 **Half price**

5 is not under 5, so it isn't free; it is under 12, so the elif runs: half price.

8 **for i in range(5):; for i in range(1, 6):; for i in range(0, 5):**

range(5) and range(0, 5) give 0 to 4; range(1, 6) gives 1 to 5: five values each.

9

```
a = int(input("First mark? "))
b = int(input("Second mark? "))
if a > b:
    print(a)
else:
    print(b)
```

One comparison decides which mark to print.

10 **False**

Comparing needs ==: if score == 10:. A single = stores a value and is a SyntaxError inside an if.

11 **Top to bottom: 2, 4, 1, 3**

The message tells you the kind of error and the line.